

Nick DiSanto

<https://nickdisanto.github.io>

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Education

Vanderbilt University

Jan. 2025 – Present

PhD in Computer Science

GPA: 4.0/4.0.

California Baptist University

Aug. 2019 – Apr. 2023

B.S. in Computer Science, *summa cum laude*

GPA: 4.0/4.0. Ranked #1 in graduating class.

Experience

Vanderbilt Medical Image Computing Lab

Jan. 2025 – Present

Graduate Student Researcher

- Developing deep learning methods for image reconstruction and restoration
- Designing registration and volumetric inpainting models for high-resolution 3D OCT data, addressing motion-induced slice corruption

GE Vernova

Jan. 2024 – Jan. 2025

Software Engineer

- Led the wildfire forecasting initiative, including the end-to-end training, visualization, and productization of ML pipelines with power grid applications
- Presented results of wildfire risk forecasting analytic to the C-Suite, helping to secure significant funding for further integration
- Honored as a 2024 GE Vernova Changemaker for co-leading a company-wide initiative to integrate GenAI into GEV Electrification Software products

CalBaptist ML & NLP Lab

Jan. 2022 – Dec. 2023

Lead Undergraduate Researcher

- Built representation models to find empirical patterns in unstructured online data, challenging the necessity of complex algorithms for natural language pattern recognition
- Proposed a novel data preprocessing approach that introduces noise and helps medical imaging models better perceive their environments, increasing accessibility for low-income communities
- Produced analyses and applications for three papers, leading the team from inception to execution

Sirch

Sep. 2023 – Dec. 2023

NLP Consultant

- Built models that perform Query Auto-Completion (QAC), information retrieval, and NLU
- Scaled product to efficiently handle increasing foot traffic while maintaining a personalized UI

General Electric

June 2022 – Aug. 2023

Software Engineering Intern

- Led a database optimization initiative that saves thousands of dollars per year
- Developed and optimized microservices for large-scale data ingestion applications

Keysight Technologies

June 2021 – Aug. 2021

Software Engineering Intern

- Created a web platform and built applications to enable clients to customize their products
- Performed data analysis to help eliminate 40% of product-option configurations

Publications & Preprints	<i>LOTUS: Latent Outpainting Diffusion Model for Three-Dimensional Ultrasound Stitching.</i> X Yao, R Yu, N DiSanto, E Aghdam, K Oguine, D Lu, A Lou, J Wang, D Hu, G Arenas, B Oguz, A Pouch, N Schwartz, B Byram, I Oguz. <i>Medical Imaging with Deep Learning (MIDL)</i>, 2025. [pdf] [code]	
	<i>Spatial Analysis of Social Media’s Proxies for Human Emotion and Cognition.</i> A Corso, N DiSanto, N Corso, E Lee. <i>International Conference on Information</i>, 2024.	
	<i>Transcending the Attention Paradigm: Representation Learning from Geospatial Social Media Data.</i> N DiSanto, A Corso, B Sanders, G Harding. <i>arXiv preprint arXiv:2310.05378</i>, 2023. [pdf] [code]	
	<i>Leveraging Contextual Data Augmentation for Generalizable Melanoma Detection.</i> N DiSanto, G Harding, E Martinez, B Sanders. <i>arXiv preprint arXiv:2212.05116</i>, 2022. [pdf] [code]	
	<i>Beyond Interpretable Benchmarks: Contextual Learning through Cognitive and Multimodal Perception.</i> N DiSanto. <i>arXiv preprint arXiv:2304.00002</i>, 2022. [pdf]	
Teaching Experience	EGR121: Intro to C++ , <i>Teaching Assistant</i>	SP 2023
	EGR225: Discrete Structures , <i>Teaching Assistant</i>	FA 2022
	CSC312: Algorithms , <i>Tutor</i>	SP/FA 2022
	EGR329: Computer Architecture , <i>Tutor</i>	SP/FA 2022
	PHY201: Physics for Engineers , <i>Teaching Assistant</i>	SP/FA 2021
Awards & Honors	2024 GE Vernova Changemaker (selected out of 1200 nominees)	Nov. 2024
	CS Outstanding Student Award (ranked #1/40)	Apr. 2023
	Inducted into Alpha Chi Honor Society (top 10%)	Apr. 2022
	Physics Performance & Leadership Award	Dec. 2021
	President’s List (every semester of undergraduate study)	FA 2019 – SP 2023
	Trustee Merit Scholarship	FA 2019 – SP 2023
	Engineering Excellence Scholarship	SP 2021 - SP 2023
Leadership & Outreach	Founder and President of CBU Chess Club.	
	Member of CBU’s Association for Computing Machinery (ACM) Chapter.	
	Lead trumpet and section leader in 4 competitive CBU musical ensembles.	
	Set up a newly-opened local high school’s computer lab.	
	Performed STAR test data analysis for an underresourced elementary school during COVID-19 pandemic.	